

Neural networks and PDE

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Outline

- 1 Motivation
- 2 Connection to PDE
- 3 Consequences

Motivation (What is machine learning)

Three types of problems

- Supervised Learning
 - ▶ Learning with a teacher
 - ▶ **Ex:** Regression / Classification

Motivation (What is machine learning)

Three types of problems

- Supervised Learning
 - ▶ Learning with a teacher
 - ▶ **Ex:** Regression / Classification
- Unsupervised Learning
 - ▶ Learning representations
 - ▶ **Ex:** Density estimation, dimensionality reduction, etc.
- Reinforcement Learning
 - ▶ Learning with a critic
 - ▶ **Ex:** Optimal control

Supervised Learning

- Classification

- ▶ Image classification: x -image, y -class. Could be object identification like saying 'this is the image of a cat'.
- ▶ Text classification: Given a snippet of text, what is its subject?

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- Regression

- ▶ What is the weight of a person given the height? x -height, y -length.
- ▶ Object location: Given that you have an image with a ball in it, where in the image is the ball.

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- Call a set of hypothesis $\mathcal{H}$, the hypothesis space.

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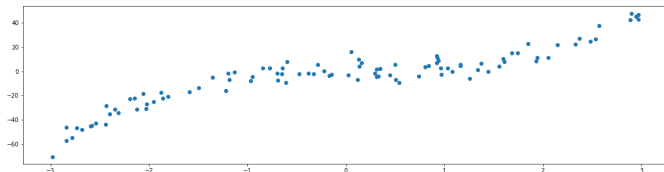
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- We cannot find h_D^* in general. Instead we try to find $h \in \mathcal{H}$ such that $R_{emp,D}(h)$ is as small as possible,

$$R_{emp,D}(h_D^*) \leq R_{emp,D}(h), \quad \textbf{Empirical Risk Min}$$

Simple example

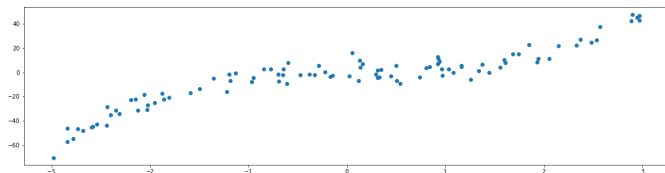


- Let $\mathcal{H}$ be functions of the form

$$h(x) = \sum_{i=1}^M v_i \sigma(w_i \cdot x + b_i)$$

for parameters v_i, w_i, b_i . $\sigma(x) = \frac{1}{1+e^{-x}}$

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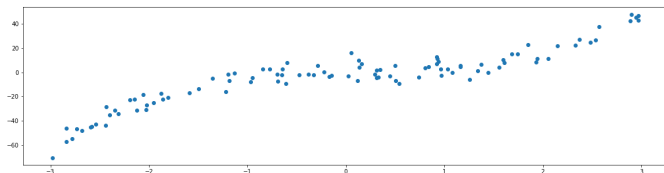
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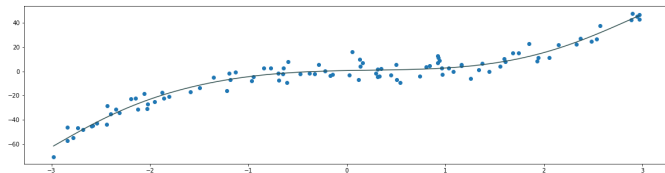
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- Goal: find h_D^* that minimizes

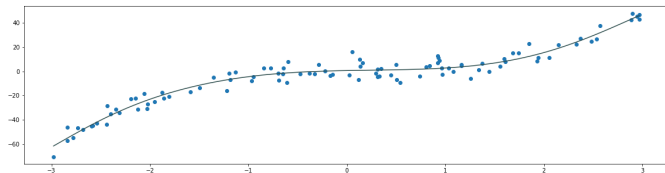
$$R_{emp,D}(h) = \frac{1}{N} \sum_{i=1}^N (h(x_i) - y_i)^2$$

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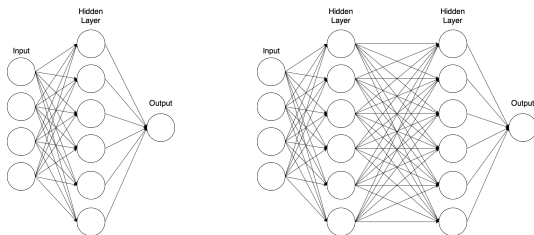


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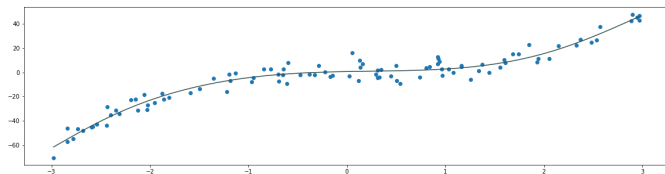
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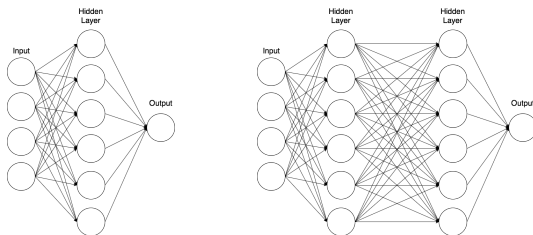
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$$h(x) = \sigma(W^2(\sigma(W^1x + B^1)) + B^2)$$

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- D_i is a subsampled set of D at each time step i , Δn is step-length.
- Called Stochastic Gradient Descent (SGD), or Robins-Monro stochastic approximation.

What are the dynamics of W_i ?

- $\nabla_W R_{emp,D_i}(h_W)$ is an unbiased estimate of true gradient $\nabla_W R_{emp,D}(h_W)$.

$$\Delta W_i = -\nabla_W R_{emp,D}(h_{W_i})\Delta n + (\nabla_W R_{emp,D}(h_{W_i}) - \nabla_W R_{emp,D_i}(h_{W_i}))\Delta n$$

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Observations

- If $\Delta n \rightarrow 0$ then we regain standard gradient flow.
- It is unclear what Σ actually is
- The density of W_t solves a Fokker-Planck equation.

Fokker-Planck

The density of

$$dW_t = -\nabla_W R_{emp,D}(h_{W_t})dt + \sqrt{\Delta n \Sigma(W_t)}dB_t$$

solves the Fokker planck equation (where $V(W) = R_{emp,D}(h_W)$)

$$\dot{\rho} = \nabla \cdot \left(\rho \nabla V + \frac{\Delta n}{2} \nabla \cdot (\Sigma \rho) \right)$$

Remember: W is high dimensional.

Gradient flow

- If $\Sigma = \sigma I$, $\Delta_n = \alpha$ and V is confining then the SDE becomes the stochastic gradient flow equation on the potential V . The corresponding Fokker planck equation.

$$\frac{\partial \rho}{\partial t} = \nabla \cdot \left(\rho \nabla V + \frac{\sigma \alpha}{2} \nabla \rho \right),$$

Has the following stationary solution

$$\rho = e^{-\frac{2}{\sigma \alpha} V}$$

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- Consider the transformation,

$$\rho_1 = e^{\frac{2}{\sigma \alpha} V} \rho,$$

- Multiply by a compactly supported test function $\varphi \in C_0^\infty$, no time dependence, then for $d\mu = e^{-\frac{2}{\sigma \alpha} V} dx$,

$$\int \frac{\partial \rho_1}{\partial t} \varphi d\mu = \int \nabla \cdot \left(\frac{\sigma \alpha}{2} e^{-\frac{2}{\sigma \alpha} V} \nabla \rho_1 \right) \varphi dx,$$

Gradient flow

- We can perform the integration by parts on the right hand side and get,

$$\int \frac{\partial \rho_1}{\partial t} \varphi d\mu = - \int \frac{\sigma \alpha}{2} \nabla \rho_1 \cdot \nabla \varphi d\mu,$$

- Rescaling the time variable leads to a heat equation w.r.t. the measure $d\mu$

$$\int \frac{\partial \rho_1}{\partial t} \varphi d\mu = - \int \nabla \rho_1 \cdot \nabla \varphi d\mu,$$

Conclusion: Our stochastic gradient flow on f gives rise to a gradient flow of the Dirichlet energy

$$E(\rho) = \frac{1}{2} \int |\nabla \rho|^2 d\mu,$$

in L^2_μ .

Consequences

- The non-convex optimization problem becomes convex in the space of distributions.
- We obtain very good tail bounds on the density.
- It becomes easier to study stability problems, for instance what happens in the infinite layer limit.
- We obtain a lot of tools to study different types of regularizers and other first order optimization methods.
- The better the estimate of the Poincaré inequality related to the measure μ the better control over convergence rate to the limit distribution.

Further reading



B. Avelin, K. Nyström,
Neural ODE as the Deep Limit of ResNets. <https://arxiv.org/abs/1906.12183>